

Transformation of calcite to fluorapatite at room temperature: Impact of initial phosphate and fluoride levels

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Abstract

Geogenic fluoride (F) contamination in groundwater is a severe problem affecting millions of people across the world. For many F-contaminated aquifers, the solution saturation state is controlled by calcite and/or dolomite solubility. Recently, apatite-based *ex situ* F-immobilization techniques have gained attention as they minimize the release of secondary pollutants, otherwise attributed to conventional sorptive F-removal methods. Likewise, addition of phosphate (PO_4) to F-contaminated subsurface environments could potentially facilitate immobilization of fluoride in sparingly soluble apatites. The success of these *ex situ* or *in situ* interventions would rely on careful understanding of the mechanisms behind F immobilization and the long-term stability of these immobilized forms. In this study, the impact of PO_4 on the extent and forms of F uptake in the presence of calcite under conditions supersaturated with respect to fluorapatite ($\text{Ca}_5(\text{PO}_4)_3\text{F}$; FA) were investigated. A year-long series of batch equilibration experiments with varying degrees of initial supersaturation, involving combinations of F (0.1–0.42 mM), PO_4 (0.7–1 mM), and calcite (0.1–4 g·L^{−1}), were conducted at room temperature. Total dissolved concentrations of F, Ca, and PO_4 were analyzed with time. Changes in solution saturation state were evaluated within a thermodynamic modeling framework comprising an updated database of relevant aqueous and mineral solubility reactions. Additionally, surface complexation modeling was used to interpret F uptake on calcite. Results from the solution chemistry indicated that the system responded to relative kinetics of dissolution of calcite and precipitation of FA or fluoridated HA. System pH and Ca concentrations in the presence of phosphate were controlled by rapid calcite dissolution, which facilitated F uptake by creating conditions highly supersaturated with respect to FA. Formation of FA was suggested by the observed PO_4 : F molar uptake ratio (~ 3), consistent with FA stoichiometry, and was confirmed with scanning electron microscopy-associated energy dispersive X-ray spectroscopy, X-ray diffraction, Fourier-transformed infrared spectroscopy, transmission electron microscopy-associated selected area diffraction, and X-ray photoelectron spectroscopy on solids collected at the end of the equilibration studies. Furthermore, stoichiometric PO_4 :F molar uptake was found only for systems containing concentrations of initial F, PO_4 , and calcite above certain critical levels (0.42 mM, 1 mM, and 1 g·L^{−1}, respectively). Surface complexation modeling predictions suggested minimal adsorption of F and PO_4 onto calcite for all conditions, irrespective of the initial concentrations. For sub-critical initial concentrations, F uptake probably occurred due to formation of fluoridated-hydroxyapatite (non-stoichiometric FA). Significantly, these immobilized F forms controlled the observed F and P concentrations below the respective drinking water and discharge limits for a year.

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